

SAT Hard Question Set 5

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Novel Prep

Overview

Question 1

The given expression can be rewritten as $\frac{a}{4}x^2 + \frac{b}{4}x + \frac{c}{4}$, where a , b , and c are constants. What is the value of $a + b + c$?

Given the expression

$$-16(5x - 3)^2 + 4(5x - 2)^2$$

- A. -112
- B. -56
- C. -28
- D. -7

Answer 1

Correct Answer: A

Question 2

In the given function, b is a positive constant. For every increase in the value of x by 1, the value of $f(x)$ increases by $c\%$, where $0 < c < 100$.

Which expression gives the value of c in terms of b ?

$$f(x) = 66(b)^x$$

- A. $1 + \frac{b}{100}$
- B. $b + 100$
- C. $100(b + 1)$
- D. $100(b - 1)$

Answer 2

Correct Answer: D

Question 3

If $\sin(x^\circ) = a$, which of the following must be true for all values of x ?

- A. $\cos(x^\circ) = a$
- B. $\sin(90^\circ - x^\circ) = a$
- C. $\sin(x^2)^\circ = a^2$
- D. $\cos(90^\circ - x^\circ) = a$

Answer 3

Correct Answer: D

Question 4

The function g is a quadratic function in the xy -plane. The graph of $y = g(x)$ has a vertex at $(-1, -4)$ and passes through the points $(-2, -43)$ and $(1, -160)$. What is the value of $g(0) - g(2)$?

- A. 312
- B. 355
- C. -117
- D. -203

Answer 4

Correct Answer: A

Question 5

The mass of object A is 324% of the mass of object B and the mass of object A is 7.2% of the mass of object C.

If the mass of object C is $p\%$ of the mass of object B, what is the value of $p\%$?

- A. 4.5%
- B. 4.5
- C. 45
- D. 450

Answer 5

Correct Answer: C

Question 6

In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c ?

$$\frac{1}{cx} = \frac{x}{76} + \frac{1}{c}$$

Answer 6

Final Answer: -19

Question 7

A parallel electric circuit contains four resistors with resistances of p ohms, q ohms, s ohms, and 13 ohms.

The given equation relates the resistance R , in ohms, of the circuit to the resistances of the four individual resistors it contains.

Which equation correctly expresses R in terms of p , q , and s ?

$$\frac{1}{R} = \frac{1}{p} + \frac{1}{q} + \frac{1}{s} + \frac{1}{13}$$

- A. $R = \frac{13pqs}{p + q + s + 13}$
- B. $R = \frac{13pqs}{13qs + 13ps + 13pq + pqs}$
- C. $R = \frac{13qs + 13ps + 13pq + pqs}{13pqs}$
- D. $R = p + q + s + 13$

Answer 7

Correct Answer: B

Question 8

In the given equation, b is an integer. The equation has no real solutions. What is the least possible value of b ?

$$-x^2 + bx - 169 = 0$$

- A. 26
- B. 25
- C. -26
- D. -25

Answer 8

Correct Answer: D

Question 9

The exponential function f is defined by $f(x) = ab^x$, where a and b are positive constants.

If $f(n-1) = f(n) + \frac{82}{100}f(n-1)$, where n is a constant, what is the value of b ?

- A. 1
- B. 0.82
- C. 0.18
- D. 10

Answer 9

Correct Answer: C

Question 10

A rectangle is inscribed in a circle such that the length of the diagonal of the rectangle is twice the length of its shortest side.

The circumference of the circle is 114π units. What is the area, in square units, of the rectangle?

- A. $57\sqrt{2}$
- B. $57\sqrt{3}$
- C. $3,249\sqrt{2}$
- D. $3,249\sqrt{3}$

Answer 10

Correct Answer: D

Question 11

In triangle PQR , the measure of angle P is $(3x + 5)^\circ$, the measure of angle Q is $(2x + 9)^\circ$, and the measure of angle R is $(4y + 5)^\circ$.

If side QR is extended through point R to point S , and the measure of angle PRS is $(x + y)^\circ$, what is the value of $x + y$?

- A. 39
- B. 29
- C. 141
- D. 146

Answer 11

Correct Answer: A

Question 12

The manager of a gym selected a sample of 139 members at random to estimate the percentage of the gym's members that would continue to pay for a membership if the price increased.

From the survey, the manager estimates that 82% of the gym's members would continue to pay for a membership if the price increased, with an associated margin of error of 6.39%.

If the survey is repeated with a random sample of 278 members and the results are calculated in the same way, which of the following will be the most likely effect of using the larger random sample compared to the smaller random sample?

- A. The margin of error will be lower.
- B. The margin of error will be higher.
- C. The estimate of the percentage of the gym's members that would continue to pay for a membership if the price increased will be lower.
- D. The estimate of the percentage of the gym's members that would continue to pay for a membership if the price increased will be higher.

Answer 12

Correct Answer: A

Question 13

Given the expression $\sqrt[5]{119n(\sqrt[6]{119n})^2}$,

for what value of x is the expression equivalent to $(119n)^{30x}$, where $n > 1$?

A. 225

B. $\frac{8}{225}$

C. $\frac{4}{225}$

D. $\frac{2}{225}$

Answer 13

Correct Answer: D

Question 14

The mass of object A is 483% of the mass of object B, and the mass of object A is 0.069% of the mass of object C.

If the mass of object C is $p\%$ of the mass of object B, what is the value of $\frac{p}{1,000}$?

- A. 0.7%
- B. 70
- C. 70%
- D. 700

Answer 14

Correct Answer: D

Question 15

For triangle PQR and triangle STU , angles P and S each measure 30° , $QR = 16$, and $TU = 12$.

Which additional piece(s) of information is(are) sufficient to prove that triangle PQR is similar to triangle STU ?

I $\angle Q = 40^\circ$ and $\angle U = 110^\circ$

II $\frac{PR}{SU} = \frac{4}{3}$

- A. I is sufficient, but II is not.
- B. II is sufficient, but I is not.
- C. I is sufficient, and II is sufficient.
- D. Neither I nor II is sufficient.

Answer 15

Correct Answer: C

Thank You!



We appreciate your time and focus.

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